

# Linear Programming Report

Sensitivity Analysis & Duality

## Unbounded Primal / Infeasible Dual

max  $x_1 + x_2$  subject to  $x_1 - x_2 \leq 1$ : the objective can grow forever along  $x_1 = x_2$ , so the dual must be infeasible.

Source: Completes the duality status table (slides 50-54)

|                   |                                                  |
|-------------------|--------------------------------------------------|
| Problem type      | MAX, 2 variables, 1 constraints                  |
| Status            | UNBOUNDED                                        |
| Optimal Objective | –                                                |
| Dual status       | INFEASIBLE                                       |
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| Course            | Sensitivity Analysis and Duality (Dr. Islam Ali) |
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## 1. Model formulation

```

MAX  z = x1 + x2
subject to
    x1 - x2 ≤ 1      (Con1)
    x1 ≥ 0, x2 ≥ 0
  
```

### Data table

| Constraint  | x1 | x2 | Sense | RHS |
|-------------|----|----|-------|-----|
| Con1        | 1  | -1 | ≤     | 1   |
| Objective c | 1  | 1  | max   | –   |

### Definitions

- **x1**: variable 1
- **x2**: variable 2

## 2. Result

The problem is **unbounded**. Its dual is **infeasible**. An unbounded primal always has an infeasible dual (weak duality).

### Dual formulation

```

MIN  z = y_Con1
subject to
    y_Con1 ≥ 1      (x1)
    -y_Con1 ≥ 1     (x2)
    y_Con1 ≥ 0
  
```

